

RESEARCH ARTICLE

Front-End Governance of Juvenile Cybercrime in China: Platform Accountability, Risk Mitigation, and Tri-Partite Collaboration

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ABSTRACT

Juvenile cybercrime is a salient governance challenge in digitalizing societies, yet scholarship concentrates on ex post sanction and under-theorizes the interaction among platform design, the digital risk environment, and minors' developmental vulnerability. Drawing on *K.G.M. v. Meta Platforms*, this study develops a “front-end governance” framework anchored in the best-interests-of-the-child principle and articulated through three doctrinal pillars, the platform duty of risk mitigation, the regulation of harmful design and risky digital environments, and the foreseeability of risk. An empirical component, based on a cross-sectional survey in China, examines institutional support for prevention. Endorsement is broad, 92% for strengthened market regulation, 91% for cyber-law education, 84% for technical firewalls, and 83% for internet classification and digital curfews. On responsibility allocation, 76% prioritize tri-partite collaborative governance involving families, schools, and society. Multinomial logistic regression indicates occupational positioning is the only significant predictor of priority orientation; regional, income, and educational variation are non-significant once controls are applied, a null pattern we read as suggestive rather than confirmatory. The findings support a shift from ex post sanction to ex ante risk governance, organized around platform accountability, age-appropriate design, rule-of-law education, and graded content ecosystems.

1 | Introduction

Minors have become one of the most active and most readily affected groups of the internet age (Lin and Qu 2025). The digital environment has, on one hand, expanded minors' channels for information access, learning, and social interaction; on the other, through algorithmic recommendation, immersive interaction, anonymous socialization, and low-threshold exposure mechanisms, it has continually amplified their probability of encountering risk-laden content, developing behavioral deviance, and even sliding into criminal activity (L. Wang et al. 2025).

Against this backdrop, juvenile cybercrime has become a critical challenge for criminal justice systems worldwide, and the problem is particularly acute in China. Official statistics from China's Supreme People's Procuratorate (SPP) document a dramatic escalation in prosecutions for juveniles involved in facilitating information network criminal activities, rising from 236 cases in 2020 to 5474 in 2022 (SPP 2023). The typology has diversified considerably, encompassing online fraud, illegal gambling, narcotics trafficking, unauthorized system intrusions, and crimes precipitated by gaming and live-streaming platforms

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(SPC 2024). Pathological internet use also appears strongly correlated with traditional street crime. Among juveniles adjudicated for robbery, theft, and violent offenses, roughly 60% exhibited chronic internet addiction, and many attributed their criminal motivations to exposure to deviant online content (SPC 2024).

These trends show that juvenile cybercrime is no longer a problem of individual deviance alone, but rather a complex governance issue intertwined with platform design, the structure of the digital environment, and the developmental vulnerability of minors (Fassi et al. 2025). A recent California verdict marks an early judicial articulation of this logic. On March 25, 2026, a California jury in *P.F. v. Meta Platforms Inc.* (the K.G.M. action) found Meta and YouTube liable for negligent design and operation of Instagram and YouTube, respectively, and for failure to adequately warn (*P.F. v. Meta Platforms Inc.* 2026b, 2026c). The jury awarded K.G.M. \$3 million in compensatory damages, and further found by clear and convincing evidence that both defendants had acted with malice, oppression, or fraud, supporting an additional \$3 million in punitive damages (Crowell and Moring LLP 2026; *P.F. v. Meta Platforms Inc.* 2026a). Although framed as a harm-to-minors case rather than one of juvenile offending, the court's reasoning bears directly on cybercrime prevention by treating platform architecture as the source of foreseeable harm. Design features such as infinite scroll, autoplay, and intermittent reinforcement were singled out on this ground.

Scholarship, however, lags behind. The prevailing research agenda is dominated by techno-centric accounts that foreground specific tools while marginalizing how juveniles actually engage with digital environments (Dearden and Parti 2021). Three blind spots stand out. First, attention concentrates on the criminogenic potential of niche tools such as anonymizers, dark-web markets, and encrypted messaging, while routine platform affordances, including recommendation engines, attention-capture mechanisms, and frictionless social connection, receive far less scrutiny, even though these features interact most directly with adolescent development (Holt et al. 2021). Second, prevention recommendations gravitate toward content moderation and law enforcement, leaving front-end design and age-appropriate governance levers, which may be more effective for juvenile populations, comparatively underdeveloped (Dupont et al. 2024). Third, the cross-actor architecture of prevention, the joint responsibilities of platforms, families, schools, and the state, remains thinly theorized (Virgara and Whitten 2023). Together, these gaps leave the field without a coherent account of how juvenile cybercrime is co-produced by platform design and developmental vulnerability.

Building on this critique, the present study reorients the analytical lens from “which technologies make crime possible” to “how platform risks and juvenile vulnerability interact to generate cybercrime risk.” Two core questions guide the inquiry: in China, (1) what should be the priorities for preventing juvenile cybercrime, and (2) which measures can effectively prevent it?

This study contributes on three fronts. Theoretically, it draws on the platform-liability reasoning developed in the K.G.M. ruling

(a U.S. juvenile online-harm action) by analogy, integrating the platform duty of risk mitigation, harmful design and risky digital environments, and foreseeability of risk into a coherent “front-end governance” framework for juvenile-cybercrime prevention. Empirically, it draws on Chinese survey data to map how different social groups perceive and support juvenile-cybercrime prevention, providing evidence on governance consensus and institutional preferences in the Chinese context. In policy terms, it argues that prevention should move beyond *ex post* punishment toward age-graded, systemic interventions coordinated across families, schools, society, and platforms, with implications for China and other rapidly digitalizing societies.

2 | Theoretical Framework

Juvenile cybercrime is not driven by technological tools alone. It arises from the interaction of the digital environment, platform mechanisms, and the developmental vulnerability of minors (Maras et al. 2024). Researchers have identified several pathways through which these interactions translate into elevated cyber-risk for juveniles, including online socialization (Obi et al. 2025), internet addiction (Wissink et al. 2023), cyber violence (Davidson and Farr 2026), and broader cyber-deviance correlates tracked longitudinally (Virgara and Whitten 2023).¹ Prevention research must therefore work across three registers at least, the normative principles governing protection of minors, the governance logic that allocates responsibility across actors, and the behavioral and design mechanisms through which risk materializes. The remainder of this section reviews each in turn.

2.1 | Normative Foundation: The Best Interests of the Child Principle

The first foundation for preventing juvenile cybercrime is the “best interests of the child” principle. In any matter affecting children, public authorities, schools, families, judicial institutions, and platform operators must give priority to four interests, namely children's survival, development, protection, and participation (UN Committee on the Rights of the Child 2021). The principle does not impose unlimited protection on minors; rather, it recognizes that their developmental, cognitive, and risk-perception immaturity warrants treatment distinct from that of adults (UN Committee on the Rights of the Child 2013). For juvenile-cybercrime prevention, the principle has a structural implication. Mainstream cyber-governance is built around adult users and presumes capacities for autonomous judgment, informed consent, and *ex post* redress that minors do not yet fully possess. Online conditions liable to induce behavioral deviance among minors therefore cannot be treated as ordinary governance problems calibrated to adults; they fall within a regime of prioritized juvenile protection that imposes distinct, *ex ante* obligations on the actors involved.

This regime translates into a duty of differentiated, prioritized, and comprehensive protection. Minors are not interchangeable with the general population of internet users, and prevention must extend beyond punishing existing harms to safeguarding their healthy development as digital subjects. Where online conditions can produce sustained addiction, victimization, or

behavioral deviance, the law's responsibility is engaged before those conditions ripen into criminal conduct.

2.2 | Governance Logic: Front-End Governance and Multi-Actor Collaborative Governance

Once the normative basis for juvenile-cybercrime prevention is established, the next question concerns the governance logic through which protection is operationalized. Conventional crime-governance models typically center on ex post sanctions: criminal prosecution, administrative penalties, or civil compensation after an offense has occurred. Juvenile cybercrime, however, is marked by gradualism and environmental dependence (Leijse et al. 2023; Paek et al. 2023; Leukfeldt and Bekkers 2026). Reliance on legal regulation only after conduct has occurred neither severs the risk chain in time nor adequately responds to the developmental needs of minors in the digital age.

This study advances “front-end governance” as the core institutional logic for preventing juvenile cybercrime. By front-end governance we mean the identification, constraint, and reconstruction of risk environments, platform mechanisms, and behavioral cues upstream of juvenile online deviance, with the aim of interrupting the trajectory from cyber harm to cyber deviance to cybercrime before harm is realized (Chhabra et al. 2025). The *P.F. v. Meta Platforms* case offers a comparative entry point for this logic, treating platform design rather than third-party content as a locus of preventive duty toward minors (*P.F. v. Meta Platforms Inc.* 2025). Although the verdict carries no direct authority outside U.S. tort law, the relevant question on a comparative reading is not whether the platform directly instigated criminal conduct, but whether design generated foreseeable risk that the operator was in a position to mitigate.

Complementing front-end governance is the logic of multi-actor collaborative governance. Single-actor protection cannot match the complexity of minors' developmental environments or the cross-domain nature of rights infringement, and the response is an integrated, closed-loop system in which multiple actors operate with delineated responsibilities and complementary capacities (Wulczyn et al. 2010). In cyberspace, families bear day-to-day supervision and value transmission. Schools deliver digital-literacy education and normative socialization. The state establishes the regulatory framework and public-governance standards. Platforms, through their control over technology, data, and contextual environments, function as indispensable nodes in risk identification and front-end intervention (Pham et al. 2024; Theopilus et al. 2025; Livingstone et al. 2026).

2.3 | Mechanism Explanations. Platform Duty of Risk Mitigation, Harmful Design and Risky Digital Environments, and Foreseeability of Risk

Within the front-end governance framework, further explanation is needed for why platforms, regulators, and other governance actors bear specific duties for the prevention of juvenile cybercrime, and on what mechanism-based grounds such duties are constituted. This study contends that three doctrines, the platform duty of risk mitigation, harmful design and risky digital environments, and foreseeability of risk, supply mechanism-based justifications, addressing, respectively, who

bears responsibility, how risk is generated, and why such duties are warranted.

2.3.1 | Platform Duty of Risk Mitigation

The doctrine of the platform duty of risk mitigation addresses the central question of why platforms cannot defend themselves merely as “neutral technology providers” or “third-party intermediaries.” Conventional internet governance has often distinguished platforms by reference to whether content originates from third parties, classifying platforms as informational intermediaries and thereby attenuating their responsibility for the consequences of user behavior. The *K.G.M. jury's* reasoning illustrates a useful analytical shift from “content neutrality” toward “platform design and operational risk,” treating certain design features as themselves capable of producing foreseeable harm to minors (*P.F. v. Meta Platforms Inc.* 2026b, 2026c). On this framing, the platform's functional design is itself the object of legal evaluation, insofar as it amplifies the probability that minors will be exposed to danger, addictive use, manipulation, or victimization.

Extending this logic to juvenile-cybercrime prevention, platforms can be said to bear an *ex ante* risk-control duty with respect to juvenile-cybercrime risk. Platforms do not passively accept risk; rather, they substantially determine what content minors encounter, how frequently it is pushed to them, how they participate in virtual interaction, and whether they can readily disengage or obtain protection (National Academies of Sciences et al. 2024). Platform responsibility, on this view, should not stop at deleting unlawful content *ex post* or cooperating with law enforcement, but should extend to risk identification, content classification, behavioral warnings, default safety settings, intervention against anomalous use, and the provision of guardian-support tools. Platforms occupy this front-end position because they are uniquely positioned to perform these functions at the point of interaction and at scale.

2.3.2 | Harmful Design and Risky Digital Environments

The doctrine of harmful design and risky digital environments explains how juvenile-cybercrime risk is actually generated. Rather than treating cybercriminal conduct as an isolated lapse of individual will or an episodic deviation, the doctrine emphasizes that online risk is typically a structural outcome accumulated within specific digital environments (F. Wang and Dickinson 2026). Although Wang and Dickinson develop this argument primarily through analysis of adult offenders on dark-web fraud forums, the affordances logic translates to juvenile contexts where platform design similarly shapes exposure patterns. Harm arises not only from particular pieces of content but, more fundamentally, from the overall behavioral environment that platforms construct through interface design, interaction feedback, recommendation logic, and attention mechanisms (Fournier et al. 2026). “Infinite scroll” illustrates the point. Even where it does no more than prolong consumption time, the combination of repeated reinforcement, intermittent reward, and erosion of decisional control can gradually push minors into compulsive engagement, vulnerability to manipulation, and exposure to deviance-inducing contexts.

The doctrine of the foreseeability of risk explains why platforms and other governance actors bear preventive duties even before harm has materialized. Conventional liability logic attaches responsibility only after injury has occurred and a causal link is established. Foreseeability supplies the legal-theoretical anchor that lets duties attach earlier, on the basis of risk rather than realized harm. In the K.G.M. action, for example, the jury expressly found that Meta knew, or in reasonable circumstances should have known, that the design and operation of Instagram were dangerous when used by minors in reasonably foreseeable ways. Although the finding has no direct legal authority outside U.S. tort law, the jury's reasoning illustrates how the duty owed to minors can be framed in concrete, foreseeable, and mitigable terms rather than as an abstract or contingent obligation.

The same logic applies to juvenile-cybercrime prevention. Because minors' cognitive capacities, impulse control, and risk-recognition abilities are not yet mature, they are more readily affected by negative influences in the online environment (Ciudad-Fernández et al. 2024). Where platforms, families, schools, or regulators can reasonably foresee that a particular online context is likely to induce criminal conduct in minors, or to render them more susceptible to becoming victims of cybercrime such as internet addiction, cyber violence, online pornography, or cyber fraud, they should bear preventive duties commensurate with the risk involved.

3 | Method

A normative framework of this kind only does institutional work if it tracks how relevant social actors actually understand responsibility and prevention. The empirical part of this study therefore turns to whether the front-end governance framework has a social-consensus base in the Chinese context, drawing on a survey of Chinese judicial practitioners and members of the general public ($N = 1197$), with associations examined through chi-square tests and multinomial logistic regression. The survey is not designed to measure the behavioral mechanisms or etiological pathways of juvenile cybercrime. It examines how the two groups perceive, prioritize, and endorse governance focal points, responsibility allocations, and prevention measures, and where their views diverge.

3.1 | Sample and Data Collection

This study employed convenience sampling, with targeted recruitment across two participant groups, namely Chinese judicial practitioners (such as procuratorate and court personnel, police officers, correctional staff, and legal professionals) and the general public. Between 2022 and 2023, 1860 questionnaires were distributed and 1260 were retrieved, yielding a retrieval rate of 67.74%. Recruitment of judicial practitioners was concentrated in Western and Central institutions accessible to the research team. No procuratorate or court personnel were sampled from Eastern China. Region and occupation are therefore not orthogonal in the present sample, the multinomial regression coefficients for region and occupation should therefore be interpreted with caution.

Three exclusion criteria were applied. The first removed surveys with more than 10% missing data on core demographic variables such as occupation or region. The second removed logically inconsistent responses, for example simultaneously selecting "no income" and "annual income > 150,000 RMB." The third removed response sets indicating careless responding, such as identical responses across five consecutive items. After screening, 63 cases were excluded, yielding a final analytical sample of 1197 valid cases.

Sample characteristics are reported in Table 1. The two participant groups were roughly balanced, with judicial practitioners just under half of respondents and members of the general public slightly over half (Table 2). Three features of the sample warrant note for the analyses that follow. Respondents were concentrated in the 20 to 40 cohort, the segment with highest internet-use intensity, and most held a bachelor's degree or higher. The sample skews female, at roughly two to one. Geographic representation is heaviest in Western China at three-fifths of respondents, with Eastern China at about one-quarter, reflecting the recruitment pattern noted above.

This study does not aim to construct a nationally representative sample. Its objective is to identify patterns of governance perception, responsibility allocation, and institutional support relevant to juvenile-cybercrime prevention. The sample is therefore best understood as an exploratory governance-perception sample, suitable for analyzing how different social groups configure their preferences within the front-end governance framework, but not appropriate for direct extrapolation to attitude distributions in the Chinese population at large. The questionnaire was developed by the authors on the basis of existing scholarship on juvenile cybercrime prevention and refined through expert review and pilot testing.

3.2 | Hypotheses

Drawing on the front-end governance framework developed in Section 2 and the occupational, regional, and sociodemographic structure of the sample described in Section 3.1, this study advances three sets of research hypotheses about how prevention cognitions vary across these dimensions.

Occupational Expertise Hypothesis (H1). Judicial practitioners' professional exposure to juvenile cybercrime cases and legal frameworks should produce distinct prevention orientations relative to the general public. Specifically:

H1a. *Judicial practitioners demonstrate higher likelihood of endorsing tri-partite collaborative governance compared to family-centered approaches.*

H1b. *Judicial practitioners exhibit elevated support for legal education and market regulation measures.*

Regional Development Hypothesis (H2). Geographic variation in economic development, digital infrastructure, and governance capacity generates regional differences in prevention cognitions. Specifically:

H2a. *Eastern region respondents show greater endorsement of technical measures and internet classification systems.*

TABLE 1 | Sample demographic characteristics.

Variable	Category	<i>n</i>	%
Occupation	Procuratorate/court personnel	493	41.19
	Police/corrections/legal	70	5.85
	General public	634	52.97
Region	Western China	730	60.99
	Eastern China	289	24.14
	Central China	91	7.60
	Other regions	87	7.27
Age	Under 20 years	110	9.19
	20–40 years	1001	83.63
	Over 40 years	86	7.18
Annual income (RMB)	20,000–100,000	378	31.58
	100,000–150,000	189	15.79
	Over 150,000	95	7.94
	No income	535	44.70
Gender	Male	449	37.51
	Female	748	62.49
Education	Graduate degree	392	32.75
	Bachelor's degree	723	60.40
	Associate degree or below	82	6.85

Note: *N* = 1197. Subtotals may not sum to 100.0% due to rounding.

TABLE 2 | Sample demographics by occupational group.

Characteristic	Procuratorate/court	Police/corrections/legal	General public	Total
Gender				
Male	188 (15.71)	39 (3.26)	222 (18.55)	449 (37.51)
Female	305 (25.48)	31 (2.59)	412 (34.42)	748 (62.49)
Region				
Western China	392 (32.75)	43 (3.59)	295 (24.64)	730 (60.99)
Eastern China	0 (0.00)	1 (0.08)	288 (24.06)	289 (24.14)
Central China	41 (3.43)	26 (2.17)	24 (2.01)	91 (7.60)
Other regions	60 (5.01)	0 (0.00)	27 (2.26)	87 (7.27)
Age				
Under 20 years	1 (0.08)	2 (0.17)	107 (8.94)	110 (9.19)
20–40 years	424 (35.42)	51 (4.26)	526 (43.94)	1001 (83.63)
Over 40 years	68 (5.68)	17 (1.42)	1 (0.08)	86 (7.18)
Education				
Graduate degree	117 (9.77)	19 (1.59)	256 (21.39)	392 (32.75)
Bachelor's degree	338 (28.24)	40 (3.34)	345 (28.82)	723 (60.40)
Associate or below	38 (3.17)	11 (0.92)	33 (2.76)	82 (6.85)
Annual income (RMB)				
20,000–100,000	253 (21.14)	33 (2.76)	92 (7.69)	378 (31.58)
100,000–150,000	167 (13.95)	20 (1.67)	2 (0.17)	189 (15.79)
Over 150,000	67 (5.60)	17 (1.42)	11 (0.92)	95 (7.94)
No income	6 (0.50)	0 (0.00)	529 (44.19)	535 (44.70)
Total	493 (41.19)	70 (5.85)	634 (52.97)	1197 (100.00)

Note: *N* = 1197. Values are frequencies with percentages of the total sample in parentheses.

H2b. *Central region respondents demonstrate stronger preference for collaborative governance models.*

H2c. *Western region respondents exhibit more heterogeneous prevention orientations.*

Sociodemographic Characteristics Hypothesis (H3). Individual-level attributes shape prevention cognitions through differential exposure to digital environments and juvenile protection concerns. Specifically:

H3a. *(exploratory): The relationship between age, income, education, and multi-measure endorsement is examined without a directional prediction.*

H3b. *Female respondents demonstrate elevated support for family-involved prevention approaches.*

3.3 | Measures

3.3.1 | Dependent Variables

Two dependent variables captured respondents' prevention orientations (Table 3). Prevention Priority Cognition (Y1) measures the perceived optimal locus of prevention responsibility. Prevention Measure Endorsement (Y2) measures support for specific intervention strategies. Both derive from survey items developed through the process described in Section 3.1.

Y1 was operationalized as a single-response item with five mutually exclusive categories. For Y2, we generated five binary indicators (endorsed vs. not endorsed) from the multiple-response item. Distribution patterns reported in Table 3 are described together with the inferential analyses in Section 4.

3.3.2 | Independent Variables

Six demographic and sociostructural variables served as predictors across all models. Table 4 presents coding schemes and distributional properties. These variables capture occupational positioning (judicial practitioner vs. general public), geographic location (regional variation in digital infrastructure and governance capacity), life-course stage (age cohorts), socioeconomic status (income levels), gender, and educational attainment. All predictors were treated as categorical variables in subsequent analyses.

3.4 | Analytical Strategy

All statistical analyses were conducted using SPSS 26.0 and STATA 17.0. Statistical significance was set at $\alpha = 0.05$ (two-tailed). Given the categorical nature of the outcome variables, a three-tier analytical strategy was implemented. Descriptive statistics were first used to summarize the sample characteristics and to provide an overview of the data structure. Crosstabulations were then performed to preliminarily explore associations between independent and dependent variables.

For the inferential analyses, Chi-square tests and Cramer's V coefficients were applied to assess the strength and significance of associations among categorical variables. Subsequently, multinomial logistic regression was estimated for the multi-category dependent variable (Y1), with Y1-1 (Tri-partite Collaborative Governance) specified as the reference category and demographic characteristics entered as covariates. For the binary measure variables (Y2-1 through Y2-5), bivariate associations with occupation, region, and demographic predictors

TABLE 3 | Dependent variable operationalization and response distributions.

Variable	Item wording (abbreviated)	Categories/coding	<i>n</i>	%
Prevention priority cognition (Y1)	"Where should prevention efforts primarily focus?"			
Y1-1		Tri-partite collaborative governance	912	76.38
Y1-2		Family-centered with auxiliary support	189	15.83
Y1-3		Dual governance (family-school)	65	5.44
Y1-4		School-focused with family coordination	21	1.76
Y1-5		Other approaches	7	0.59
Prevention measure endorsement (Y2)	"Which measures can effectively prevent juvenile cybercrime?" (multiple response)			
Y2-1		Internet classification/digital curfews (1 = endorsed)	998	83.38
Y2-2		Enhanced cyber law education (1 = endorsed)	1091	91.14
Y2-3		Strengthened market regulation (1 = endorsed)	1100	91.90
Y2-4		Technical firewall implementation (1 = endorsed)	1002	83.71
Y2-5		Other measures (1 = endorsed)	125	10.44

Note: Y1 *N* = 1194. Y2 *N* = 1197.

TABLE 4 | Independent variable operationalization.

Variable	Coding scheme
X1: Occupational group	1 = procuratorate/Court personnel; 2 = police/Corrections/Legal; 3 = general public
X2: Region	1 = western China; 2 = eastern China; 3 = central China; 4 = other regions
X3: Age	1 = under 20 years; 2 = 20–40 years; 3 = over 40 years
X4: Annual income	1 = 20,000–100,000 RMB; 2 = 100,000–150,000 RMB; 3 = over 150,000 RMB; 4 = No income
X5: Gender	1 = male; 2 = female
X6: Education	1 = graduate degree; 2 = bachelor's degree; 3 = associate degree or below

TABLE 5 | Prevention measure endorsement by region.

Measure	Western China	Eastern China	Central China	Other regions	χ^2 (df = 3)	p
Y2-1: Classification/curfews	609 (83.42)	241 (83.39)	75 (82.42)	73 (83.91)	0.12	0.99
Y2-2: Law education	668 (91.51)	261 (90.31)	82 (90.11)	80 (91.95)	0.53	0.91
Y2-3: Market regulation	672 (92.05)	264 (91.35)	82 (90.11)	80 (91.95)	0.60	0.90
Y2-4: Technical firewalls	608 (83.29)	244 (84.43)	76 (83.52)	73 (83.91)	0.19	0.98

Note: N = 1197. Values are frequencies with percentages in parentheses.

were examined via Pearson chi-square tests, the results of which are reported in Tables 5–7.

Model fit was evaluated using likelihood ratio tests, Nagelkerke pseudo- R^2 , and classification accuracy. Multicollinearity diagnostics yielded variance inflation factors below the conventional threshold of 10 across all predictors. Given the exploratory nature of this study and the absence of prior effect-size estimates for these relationships in the Chinese context, no formal power analysis was conducted, a limitation we note when interpreting nonsignificant results.

4 | Results

4.1 | Prevention Measure Endorsement: Consensus and Group Variation

4.1.1 | Aggregate Endorsement Patterns

Table 8 reports endorsement rates for five prevention measures. Four interventions drew support from more than 80% of respondents. Online cultural market regulation (Y2-3, 91.90%) and cyber law education (Y2-2, 91.14%) received the highest endorsement; technical firewalls (Y2-4, 83.71%) and graded internet access with digital curfews (Y2-1, 83.38%) followed closely. Alternative measures (Y2-5) drew 10.44%.

Two readings follow. Respondents in this sample do not reduce juvenile cybercrime to a problem of “insufficient individual self-control” or “failures of family education,” but treat institutional, technical, and normative measures as necessary components of prevention. The over-90% support for market regulation and cyber law education is consistent with a perception of juvenile-cybercrime prevention as an institutional issue requiring joint engagement from the state, platforms, and society, and provides cognition-level evidence that aligns with, though does not by itself establish, the front-end governance framework set out above.

4.1.2 | Demographic Variation in Measure Endorsement

Chi-square tests examined whether endorsement rates varied across demographic subgroups for the four primary measures (Y2-1 through Y2-4). Variation was limited.

The first is regional homogeneity. Cross-tabulation detected no statistically significant regional differences in endorsement rates across all four primary measures (Table 5). Western China respondents endorsed market regulation at 92.05%, compared to 91.35% in Eastern China and 90.11% in Central China. Similar patterns emerged for law education, technical firewalls, and classification systems. These cross-regional patterns are consistent with broadly shared prevention concerns rather than region-specific demand structures, and do not support H2's prediction of regional variation in endorsement of technical measures.

The second is occupational differentiation. Judicial practitioners and the general public demonstrated subtle differences in endorsement patterns (Table 6). For market regulation (Y2-3), procuratorate/court personnel exhibited numerically elevated endorsement compared to the general public and police/corrections/legal professionals, though differences lacked statistical significance. Similarly, law education endorsement varied modestly across occupational groups, as did technical firewall support and classification system preferences.

Police/corrections/legal practitioners endorsed alternative measures (Y2-5) at numerically higher rates than procuratorate/court personnel and the general public, although the difference did not reach conventional significance. The pattern is directionally consistent with H1b, but limited subsample size precludes firmer inference.

The third is income-invariant patterns. Income level demonstrated no statistically significant associations with measure endorsement across all four primary interventions (Table 7).

TABLE 6 | Prevention measure endorsement by occupational group.

Measure	Procuratorate/court	Police/corrections/legal	General public	χ^2 (df = 2)	<i>p</i>
Y2-1: Classification/curfews	411 (83.37)	59 (84.29)	528 (83.28)	0.08	0.96
Y2-2: Law education	447 (90.67)	65 (92.86)	579 (91.32)	0.52	0.77
Y2-3: Market regulation	455 (92.29)	64 (91.43)	581 (91.64)	0.18	0.91
Y2-4: Technical firewalls	414 (83.98)	60 (85.71)	528 (83.28)	0.44	0.80
Y2-5: Other measures	39 (7.91)	11 (15.71)	75 (11.83)	5.89	0.053

Note: *N* = 1197. Values are frequencies with percentages in parentheses.

TABLE 7 | Prevention measure endorsement by income level.

Measure	20–100 K	100–150 K	Over 150 K	No income	χ^2 (df = 3)	<i>p</i>
Y2-1: Classification/Curfews	314 (83.07)	158 (83.60)	80 (84.21)	446 (83.36)	0.09	0.99
Y2-2: Law education	343 (90.74)	173 (91.53)	86 (90.53)	491 (91.78)	0.67	0.88
Y2-3: Market regulation	346 (91.53)	174 (92.06)	88 (92.63)	491 (91.78)	0.23	0.97
Y2-4: Technical firewalls	316 (83.60)	159 (84.13)	81 (85.26)	446 (83.36)	0.31	0.96
Y2-5: Other measures	42 (11.11)	21 (11.11)	9 (9.47)	53 (9.91)	0.48	0.92

Note: *N* = 1197. Values are frequencies with percentages in parentheses. Income categories in RMB.

TABLE 8 | Prevention measure endorsement rates.

Measure	Endorsed	Not endorsed	Endorsement rate (%)
Y2-1: Internet classification & digital curfews	998	199	83.38
Y2-2: Enhanced cyber law education	1091	106	91.14
Y2-3: Strengthened market regulation	1100	97	91.90
Y2-4: Technical firewall implementation	1002	195	83.71
Y2-5: Other measures	125	1072	10.44

Note: *N* = 1197. Multiple responses permitted. Percentages calculated independently for each measure.

High-income respondents (over 150,000 RMB) endorsed legal education at 90.53%, while no-income respondents endorsed it at 91.78%. Market regulation endorsement ranged from 91.53% (20,000–100,000 RMB) to 92.63% (over 150,000 RMB), yielding no significant variation. Similar patterns emerged for technical firewalls and classification systems. This income-invariant pattern is consistent with cross-class endorsement of core prevention measures. Because H3a was specified as exploratory and non-directional, no directional confirmation is claimed; the null finding cannot be distinguished from limited statistical power or sampling concentration in particular income brackets.

The fourth is gender and education effects. Neither gender nor educational attainment demonstrated statistically significant associations with measure endorsement. Female respondents exhibited endorsement rates statistically indistinguishable from male counterparts across all four primary measures (all $p > 0.40$). Similarly, educational attainment showed no systematic relationship with endorsement patterns (all $p > 0.50$), failing to support the educational dimension of H3a. These null findings are consistent with cross-gender and cross-education consensus on core prevention measures, but cannot rule out limited power as an alternative explanation.

Two patterns emerge. Respondents' cognitions on specific prevention measures are highly concentrated, and most demographic variables generate no significant variation. At the

level of concrete institutional instruments, front-end measures attract broadly distributed endorsement. Variation that the data do reveal lies elsewhere, in institutional understandings of who should bear principal responsibility for governance, which the next subsection examines.

4.2 | Prevention Priority Orientations

4.2.1 | Aggregate Distribution

Responses to the single-item measure “Where should prevention efforts for juvenile cybercrime primarily focus?” revealed overwhelming consensus favoring tri-partite collaborative governance involving families, schools, and society (Y1-1), with alternative orientations each garnering less than 16% endorsement (Table 9). Family-centered approaches with auxiliary support (Y1-2) constituted the second most prevalent orientation at 15.83%, while family-school dual governance (Y1-3), school-focused coordination (Y1-4), and other approaches (Y1-5) collectively accounted for less than 8% of responses. This concentration suggests substantial social agreement on collaborative prevention architectures.

Respondents in this sample do not treat juvenile-cybercrime prevention as a private family matter. They understand it as a shared responsibility across multiple actors, with platform governance, social support, and educational mechanisms read

TABLE 9 | Prevention priority orientation distribution.

Priority orientation	<i>n</i>	%
Y1-1: Tri-partite collaborative governance	912	76.38
Y1-2: Family-centered with auxiliary support	189	15.83
Y1-3: Family-school dual governance	65	5.44
Y1-4: School-focused with family coordination	21	1.76
Y1-5: Other approaches	7	0.59
Total	1194	100.00

Note: *N* = 1194. Three cases excluded due to missing responses.

as parallel governance components rather than supplements to family responsibility. Whether this distribution of cognitions tracks legal-doctrinal arguments for collaborative governance is a separate question that the cross-cutting fit between cognition and doctrine cannot itself settle.

4.2.2 | Bivariate Associations

Chi-square analyses examined associations between demographic predictors and prevention priority orientations (Table 10). Four predictors demonstrated statistically significant bivariate relationships: occupational group, income, region, and age. Effect sizes ranged from small to moderate, with occupational group and income exhibiting strongest associations. Gender and education showed no significant bivariate relationships.

Three patterns appear at the bivariate level. Police and corrections/legal professionals show the highest tri-partite governance endorsement, while the general public shows elevated family-centered preferences relative to procuratorate/court personnel. Income tracks governance preferences in the same direction. High-income respondents lean toward collaborative-governance preferences. No-income respondents favor family-centered approaches at elevated rates. Regionally, Central China shows the strongest tri-partite endorsement and the lowest family-centered preferences, with Eastern China leaning toward family-centered approaches. Whether these bivariate patterns survive multivariate controls is examined in the next subsection.

4.2.3 | Multinomial Logistic Regression Results

To assess unique contributions of each predictor while controlling for covariates, we estimated multinomial logistic regression models with tri-partite collaborative governance (Y1-1) as the reference category. The full model included all six demographic predictors and converged after 10 iterations. Overall model fit reached statistical significance, though predictive capacity remained modest.

Table 11 presents relative risk ratios (exponentiated coefficients) with 95% confidence intervals for all predictor-outcome combinations. Results revealed occupational group as the sole statistically significant predictor in multivariate context. Relative to the general public (reference category), procuratorate/court personnel demonstrated significantly reduced relative risk of endorsing both family-centered approaches and school-focused prevention, supporting H1a's prediction that judicial practitioners exhibit stronger preferences for collaborative governance

TABLE 10 | Bivariate associations between demographics and prevention priorities.

Predictor	χ^2	df	<i>p</i>	Cramér's <i>V</i>
Occupational group	44.27	8	< 0.001	0.14
Income	47.58	12	< 0.001	0.13
Region	22.31	12	0.034	0.08
Age	21.90	8	0.005	0.10
Gender	3.85	4	0.426	0.06
Education	9.12	8	0.332	0.07

models. Police/corrections/legal professionals displayed similar directional effects, though coefficients did not reach conventional significance thresholds, a pattern we read with caution given the limited subsample size in this group.

Contrary to bivariate patterns, region, age, income, gender, and education demonstrated no statistically significant effects in multivariate models, failing to support H2 and H3 predictions. Specifically, no regional dummy variable achieved significance across any outcome contrast, contradicting H2's predictions regarding Central China's collaborative preferences and Eastern China's family-centered tendencies. Age cohorts showed no significant differences, income categories demonstrated no associations, and gender and education effects remained non-significant. The attenuation of bivariate associations in multivariate contexts suggests apparent demographic differences may reflect unmeasured confounding, shared variance among predictors, or insufficient statistical power.

The multinomial regression points to two implications. First, occupational position emerges as the sole stably significant predictor in this sample, suggesting that institutional-practice experience is associated with stronger preferences for collaborative governance. The front-end governance framework, in particular, finds its strongest endorsement among professional groups familiar with juvenile-cybercrime adjudication. Second, regional, income, and educational variables yielded no statistically significant effects after controls (Table 11). This null pattern is consistent with widely shared concern about juvenile online risk across demographic strata, but cannot be distinguished from sampling concentration in the Western region or limited statistical power, and we therefore treat it as suggestive rather than confirmatory.

5 | Discussion

Two patterns in the data carry weight for how the front-end governance framework should be read. The first is the cross-cutting endorsement of institutional, rather than individual, instruments of prevention. Respondents in this sample do not locate the prevention task in moral exhortation or family education alone. They locate it in platform conduct, technical infrastructure, legal socialization, and content-environment design. The second is the limited demographic structure of disagreement. Region, income, age, gender, and education do not stably differentiate prevention preferences once covariates are controlled. The cleavage that does appear runs along occupational exposure to juvenile-justice practice, suggesting that

TABLE 11 | Multinomial logistic regression predicting prevention priority orientations.

Predictor	Y1-2 versus Y1-1 (Family-centered) RRR [95% CI]	Y1-3 versus Y1-1 (Dual governance) RRR [95% CI]	Y1-4 versus Y1-1 (School-focused) RRR [95% CI]
Occupation (ref: general public)			
Procuratorate/court	0.42*** [0.26, 0.67]	0.82 [0.44, 1.53]	0.26* [0.08, 0.91]
Police/corrections/legal	0.69 [0.28, 1.71]	0.54 [0.12, 2.46]	0.16 [0.02, 1.37]
Region (ref: Western China)			
Eastern China	1.23 [0.81, 1.87]	1.02 [0.53, 1.98]	0.67 [0.20, 2.24]
Central China	0.51 [0.22, 1.18]	0.68 [0.25, 1.83]	0.89 [0.20, 4.02]
Other regions	0.87 [0.46, 1.65]	0.91 [0.35, 2.35]	1.42 [0.34, 5.87]
Age (ref: 20–40 years)			
Under 20 years	1.35 [0.75, 2.42]	1.68 [0.77, 3.67]	2.01 [0.58, 6.93]
Over 40 years	0.71 [0.35, 1.43]	0.58 [0.18, 1.84]	0.45 [0.06, 3.61]
Income (ref: no income)			
20,000–100,000 RMB	0.88 [0.54, 1.43]	0.72 [0.36, 1.44]	0.53 [0.16, 1.76]
100,000–150,000 RMB	0.76 [0.42, 1.38]	0.67 [0.28, 1.60]	0.71 [0.20, 2.56]
Over 150,000 RMB	0.58 [0.25, 1.34]	0.91 [0.34, 2.44]	0.42 [0.05, 3.38]
Gender (ref: male)			
Female	1.21 [0.82, 1.79]	1.15 [0.65, 2.04]	1.33 [0.53, 3.35]
Education (ref: associate or below)			
Bachelor's degree	0.79 [0.39, 1.61]	0.63 [0.26, 1.53]	0.88 [0.22, 3.51]
Graduate degree	0.71 [0.34, 1.51]	0.58 [0.22, 1.50]	0.75 [0.17, 3.28]
Model fit statistics			
Likelihood ratio χ^2 (df = 48)	94.52***		
Nagelkerke R^2	0.10		
McFadden R^2	0.05		
Overall correct classification	77.5%		

Note: $N = 1187$. Y1-5 (Other approaches, $n = 7$) was excluded from the multinomial regression due to sparse cell counts that would yield unstable estimates. Y1-1 (Tri-partite Collaborative Governance) serves as the reference category.

Abbreviation: RRR = relative risk ratio.

* $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$.

disagreement about prevention is less a public-attitudes problem than a problem of which institutional vantage points become decisive in design.

The remainder of this section reads each of the four mechanisms in the framework against these patterns. The aim is not to claim that the framework has been validated by the data. The sample is exploratory, the dependent variables are stated preferences, and several effects survived only at the bivariate stage. The aim is rather to ask, mechanism by mechanism, where the cognitions documented here align with the institutional moves the framework prescribes, where they pull against those moves, and where the data are silent and design must proceed on other grounds.

5.1 | Market Regulation and Platform Governance

The results show that more than 90% of respondents support strengthening the regulation of online cultural markets, and the

great majority simultaneously endorse technical safeguards and tri-partite collaborative governance. Respondents in the sample no longer treat platforms as neutral technological tools or passive informational intermediaries, but as governance nodes that connect content exposure, behavioral inducement, and risk diffusion. The cognitions documented here are consistent with the platform-responsibility logic developed in Section 2.3 from the K.G.M. case, and with empirical work showing platforms as sites of risk concentration for young users (Ciudad-Fernández et al. 2024).

Platform governance should not stop at deleting unlawful content or assisting law enforcement after the fact. It should advance toward three categories of front-end governance mechanisms. First, platforms could be expected to deploy content-filtering systems supported by artificial-intelligence analytics for the real-time identification of harmful content such as pornography, incitement to violence, and self-harm promotion (Information Commissioner's Office 2021; Livingstone et al. 2026). Such systems

would operationalize the technical-firewall concept endorsed by 84% of respondents while addressing the monitoring-capacity deficits identified in prior research (Sala et al. 2025). Second, risk-graded labeling could be developed for content potentially affecting psychological health, implementing prominent labels rather than blanket blocking. This approach respects juveniles' information-access needs while enhancing risk cognition through technical means. Third, anomalous-behavior detection mechanisms warrant exploration, including models that draw on login patterns, consumption frequency, and social-relational signals to identify risks that may trigger intervention, extending protection from content filtering to behavioral safeguarding.

Platform governance, however, does not mean shifting all responsibility onto platforms themselves. As the results section showed, respondents strongly support tri-partite collaborative governance, indicating that platform responsibility must be embedded within a broader institutional framework. Platforms function as crucial intermediaries within front-end governance, but they cannot substitute for parental supervision, school-based education, or state regulation. Effective market regulation requires that platform responsibility be defined alongside state-set standards, school-based education, and family-based supervision, generating a coordinated governance configuration.

5.2 | Technical Infrastructure and Age-Appropriate Design

In this study, more than 80% of respondents endorse “implementing technical firewalls” and “internet classification/digital curfews,” indicating broad endorsement of technical safeguards as a tool of juvenile-cybercrime prevention. Respondents in this sample treat juvenile online protection not only as a matter of moral exhortation or legal admonition, but as a question of technical infrastructure for front-end governance. The pattern aligns with the harmful-design and risky-digital-environments doctrine. If risk accumulates within particular digital environments, prevention cannot rely on individual self-discipline alone, and the environment itself must be reconstructed through technical means.

Technical safeguards, however, do not entail constructing a fully sealed environment, an approach inconsistent with social reality and contrary to the developmental needs of minors (Hammond et al. 2023). Online risk is not equivalent to actual harm. A conservative protectionism deprives minors of opportunities to explore, adapt, and cope, and may also displace their online behavior into environments more hazardous than those subject to family or school supervision (Pan et al. 2024). The protection of minors online therefore favors guidance over prohibition. A coordinated emphasis on safety and development is more consistent with the best-interests principle, reducing the negative influences minors encounter online while preserving the developmental opportunities afforded by digital media.

In this sense, technical-infrastructure development should, on the one hand, take the form of age-appropriate design. Platforms and device manufacturers should establish differentiated rules across interface presentation, accessible content, usage duration,

interactive functions, and consumption permissions, calibrated to the developmental stage of the minor (Information Commissioner's Office 2021). On the other hand, governments should establish unified technical-governance standards to address cross-platform variation in protective intensity and implementation. Three categories merit priority. Information-filtering standards should define identification thresholds for harmful content types. Anti-addiction technical specifications should stipulate usage-duration calculation methods and mandatory rest triggers. Parental-control tool function inventories should clarify remote monitoring, consumption limits, and content-blocking capabilities. Unified standards address protection gaps arising from cross-platform variation in safeguard intensity, ensuring comprehensive coverage rather than selective enforcement that would create exploitable vulnerabilities.

5.3 | Legal Education and Normative Socialization

The results show that more than 90% of respondents support “strengthening cyber rule-of-law education,” which together with market regulation constitutes the most strongly supported governance measure. The finding indicates that platform-level restrictions and technical filtering alone are insufficient. Minors must also develop, through education, a working understanding of rule boundaries, the consequences of risky conduct, and digital responsibility. The finding resonates with the institutional logic of foreseeability theory. If foreseeability means that platforms, families, schools, and regulators must perform preventive duties before risks materialize, then in practice an important component of these duties is to help minors form sound conceptions of online norms, enabling them to identify high-risk situations, understand the consequences of unlawful conduct, and connect virtual behavior with real-world responsibility. Many juvenile-cybercrime acts arise not from a mature criminal intention, but from a weak normative awareness, insufficient risk perception, and blurred boundaries between virtual and real (Payne et al. 2020). The core objective of rule-of-law education is therefore to facilitate the normative transition from technical user to digital citizen (Payne et al. 2020; Wissink et al. 2023; Pham et al. 2024).

Effective legal education would address three objectives. First, instruction should move away from “network-avoidance education” toward an “instrumental rationality” orientation that normalizes technology use while teaching responsible application. This addresses the technical-ignorance-to-risk pathway and aligns with the 84% of respondents endorsing technical safeguards. Second, legal and ethical education should be delivered together through pedagogical models combining legal interpretation, case-based warnings, and behavioral guidance. Curricula that integrate developmental-psychology insights into rule-of-law instruction help juveniles identify high-risk behaviors and can be complemented by individualized intervention plans, including family therapy that addresses chaotic home environments and reduces emotionally driven cyber-offending. Third, virtual-reality cognitive-balance guidance should address the reality alienation produced by digital immersion, building critical engagement with online society while incentivizing real-world experience and reinforcing the principle that virtual

behavior has boundaries and real-world responsibility requires accountability.

5.4 | Age-Graded Content Ecosystems and Graduated Access

Beyond platform responsibility and technical infrastructure, 83% of respondents endorse targeted website construction and classification measures, suggesting broad recognition of age-graded digital-environment design as a relevant component of juvenile-cybercrime prevention. Juvenile online protection should not rely solely on ex post deletion and provisional blocking; rather, through long-term institutional arrangements, it should establish a graded content ecosystem aligned with minors' age, developmental stage, and risk tolerance (Theopilus et al. 2025).

From the perspective of front-end governance, the core of a graded content ecosystem is not the construction of an absolutely closed environment, but the use of scientific classification and tiered access controls to expose minors to digital risks that are foreseeable, identifiable, and manageable. In terms of concrete institutional design, the study proposes three directions for the development of a graded content ecosystem. First, establishing a dual-dimension classification standard along both an age axis and a content-risk axis. The age axis should distinguish among preschool children, primary-school students, junior-high students, and senior-high students; the content-risk axis should distinguish among general information, information potentially affecting psychological health, and clearly harmful information. Second, an implementation mechanism should combine government guidance, industry self-regulation, and family participation. Regulators set baseline rules. Industry associations refine implementation standards. Platforms apply specific labels and content adaptations. Families perform individualized configuration through unified query and filtering services. Third, content-identification and classification technology should advance through natural-language processing and computer vision, enhancing the automation and timeliness of content grading and reducing the governance vacuum produced by manual-review lag.

5.5 | Limitations and Future Research Directions

Six limitations bound the inferences this study can support. First, the cross-sectional design precludes causal inference. We have read the occupational difference as evidence of professional socialization, but the data cannot distinguish socialization from selection. Individuals with pre-existing collaborative orientations may sort into judicial professions. Longitudinal designs tracking attitude change after occupational entry would help distinguish the two.

Second, the sample's geographic concentration in Western China limits generalizability and complicates the regional null finding. Because no judicial practitioners were recruited from Eastern China, region and occupation are not orthogonal in the analytic sample. The regional null in the multinomial regression should therefore be read partly as a feature of the recruitment design rather than as substantive cross-regional uniformity. Even where statistical controls show no regional effect,

unmeasured regional characteristics may shape how respondents form prevention preferences. Nationally representative replication would test whether the observed patterns hold across China's diverse regions. Comparative research across jurisdictions with different internet-governance traditions would test whether the consensus patterns documented here are specific to the Chinese institutional context.

Third, the questionnaire data measure respondents' support for governance priorities and policy instruments, not juvenile-cybercrime behavior itself. The study can therefore document cognitive support for the front-end governance framework, but cannot show how internet addiction, cyber violence, online pornography, and online socialization translate into criminal conduct at the individual level. Future research combining judicial case data, platform behavioral records, juvenile interviews, and longitudinal tracking would let researchers examine the behavioral mechanisms underlying juvenile cybercrime more directly.

Fourth, recruitment proceeded by convenience and produced a sample skewed toward female respondents at roughly two to one. Gender did not emerge as a stable predictor in the multinomial regression, but the imbalance constrains how confidently any null finding on gender can be carried beyond the sample. Probability sampling with quota controls on gender and occupation would let future work test the same hypotheses on a more demographically balanced base.

Fifth, although the study uses the K.G.M. case as a comparative-law entry point for constructing the theoretical framework, it does not develop a systematic cross-jurisdictional comparison. Future research could compare institutional models across jurisdictions concerning platform responsibility, juvenile protection, and front-end governance, thereby testing whether the analytical framework proposed here has broader applicability.

Sixth, and most fundamentally, the study's empirical core captures the governance preferences of adult judicial practitioners and members of the public on juvenile cybercrime prevention, not juvenile psychology, online behavior, or self-reported digital experience. Claims in Section 2 about how online environments shape juvenile cognition and conduct should accordingly be read as the developmental scaffolding behind respondents' policy beliefs, not as fresh empirical findings about juveniles. This design choice is deliberate. Front-end governance design, the paper's central concern, hinges on a defensible reading of expert and public consensus about which institutional arrangements are workable and legitimate, a question properly addressed by surveying the actors who design, operate, and consent to those institutions, rather than the regulated population. Future work drawing on juvenile-respondent panels, school-administered surveys, or anonymized behavioral-trace data would help test the developmental premises underlying our governance recommendations and triangulate adult policy preferences with juvenile lived experience.

6 | Conclusion

Juvenile cybercrime is produced upstream of the criminal process. By the time the minor stands accused, the platform's

design has structured the environment, the foreseeable risks have materialized, and prevention is no longer possible. This article has argued that the response, particularly in Chinese law, should be calibrated to that fact. Preventive duty must apply where the risk environment is produced, in platform design; it must operate prospectively, before harm rather than after; and it must be distributed across family, school, and society. The survey results show that this tri-partite allocation is already an institutional consensus. The doctrine requires reading the best-interests-of-the-child principle as a positive structural obligation, not merely as an interpretive guide to disputes already in progress.

This study makes three theoretical contributions. First, it re-locates preventive responsibility from reactive criminal-justice response to front-end, design-stage governance, situating the duty to prevent at the architectural layer of digital products and institutional arrangements. Second, it operationalizes a tri-partite allocation framework and shows that this allocation enjoys structural consensus across judicial practitioners and lay respondents, giving empirical grounding to a collaborative-governance thesis previously argued on normative grounds. Third, it identifies a consensus-divergence pattern in Chinese governance preferences, broad agreement on prevention principles alongside profession-specific divergence on technical and legal-educational instruments, refining accounts that treat such preferences as either monolithic or wholly fragmented.

Three practical contributions follow. First, the study identifies platform-side technical measures, namely AI-supported content filtering, age-appropriate interface design, and behavioral-anomaly detection, that command broad stakeholder endorsement and can therefore serve as regulatory baselines rather than aspirational standards. Second, it supplies a calibration baseline for graduated-access regulation by mapping which restrictions enjoy public legitimacy and which prompt professional reservations, offering regulators a defensible starting point for age-tiered design rules. Third, it yields concrete design points for legal-education and normative-socialization interventions in school and family settings, anchored in the divergence between practitioners and lay respondents, so that interventions can be sequenced and audience-targeted rather than uniformly delivered.

This article cannot show that front-end governance, once instituted, reduces the rate at which minors enter the criminal justice system. That question is empirical, and the longitudinal, behavioral, and comparative data needed to answer it lie beyond the present design. The argument advanced here is for the doctrinal architecture within which the question becomes tractable.

Author Contributions

Zhengfa Zi: conceptualization–provided the research theme and core ideas, data curation–collected data, designed, distributed, and retrieved the questionnaire, investigation–implemented the questionnaire survey and data collection process. **Ziteng Yuan:** conceptualization–provided the manuscript writing framework, data curation–organized and collated raw data, writing – original draft–composed the initial draft of

the manuscript. **Shiqi Zhang:** writing – review and editing–revised and polished the initial draft, validation–proofread the text content and verified the accuracy of data. **Wanqiang Wu:** translation–provided core translation services for the manuscript, supervision–oversaw the overall research progress, conceptualization–offered insights on data application strategies.

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Ethics Statement

The study protocol for “Preventing Juvenile Cybercrime: An Empirical Study on Network-Induced Factors and Collaborative Governance in China” was reviewed and approved by the Institutional Review Board of Sun Yat-sen University School of Law ([Approval No. SYSU-LAW-IRB-2025-003]). All procedures involving human participants complied with the ethical standards of the institutional research committee and with the principles of the Declaration of Helsinki and Wiley’s Best Practice Guidelines on Research Integrity and Publishing Ethics. The Institutional Review Board of the School of Law, Sun Yat-sen University was only recently established in its current formal structure, which explains why the official approval letter bears a relatively recent date. However, before any data collection for this study commenced, the research protocol had already been reviewed and approved by a temporary ethics review panel appointed by the law school, and the newly constituted Board subsequently confirmed and recorded this prior approval.

Consent

All participants were informed of the purpose of the study, the voluntary nature of their participation, and their right to withdraw at any time without penalty. Prior to completing the questionnaire, participants received an information sheet describing the study and data protection arrangements. Informed consent was obtained from all individual participants before data collection. The survey was administered and stored in anonymized form; no directly identifying personal information was collected, and data are reported only in aggregate so that no individual respondent can be identified. This study does not report any identifiable personal data, images, or case-level narratives of individual participants. Consent for publication is therefore not applicable.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Endnotes

¹ The precise causal mechanisms linking these pathways to juvenile cyber-offending remain debated in the literature and are not tested by the present study, which focuses on governance preferences rather than individual-level behavioral mechanisms.

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